

Youssef MILED

Berkeley, CA, USA | *Graduating May 2026 — Available June 2026*

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EDUCATION

- UC Berkeley**, *MEng in Operations Research*, California, USA Aug 2025 – May 2026
- Coursework: Machine Learning, Agentic AI, Mathematical Programming, Stochastic Optimization.
 - Fung Institute MEng Technical Leadership Award**: Led capstone team to best project award for technical scoping and solution design.
- Centrale Lyon**, *BSc in General Engineering*, Lyon, France 2023 – 2025
- Coursework: Applied Mathematics, Computer Science, Machine Learning.
- Champollion**, *Intensive Preparatory Program — French Engineering Schools*, Grenoble, France 2021 – 2023
- Coursework: Algorithms, Data Structures, Machine Learning, C, OCaml, and Discrete Mathematics.

PROFESSIONAL EXPERIENCE

- Graduate Research Assistant**, *Satlyt*, San Francisco, USA Sep 2025 – May 2026
- Architected and shipped an edge-AI telemetry analysis system for satellites; designed retrieval and tool-calling workflows optimized for constrained compute, hitting 62.6 tok/s throughput under memory/latency budgets.
 - Built end-to-end evaluation infrastructure benchmarking latency-quality tradeoffs across prompting and retrieval strategies; achieved 7.81/10 diagnostic accuracy and +41.8% troubleshooting coverage improvement via RAG.
- ML Research Intern**, *CISPA Helmholtz Center for Information Security*, Germany May 2025 – July 2025
- Developed Cascade Unlearning: machine unlearning framework escalating in-context → LoRA → full fine-tuning by forget-set size; applied reversed self-distillation to align model versions and minimize membership inference risk.
 - Validated on SST-2, AGNews, and MIT Movies; audited privacy risk using membership inference attacks (LiRA) with TPR@1%FPR reporting; automated experiment sweeps via Docker + SLURM on OLMo 2 1B.

PROJECTS

Vulcan OmniPro Assistant

- Built and deployed a full-stack multimodal assistant (React 18, TypeScript, Tailwind CSS frontend; FastAPI + PostgreSQL backend) on Fly.io via Docker; five specialized agents orchestrated through a central router with WebSocket streaming and LRU caching for sub-second repeated query responses.
- Engineered a vision agent parsing product manual page images to generate interactive data visualizations (React components, SVG diagrams, Mermaid flowcharts) from visual content; implemented hybrid BM25 + semantic search with cross-encoder reranking.

Voice AI Freight Platform

- Built end-to-end freight brokerage platform with Next.js (TypeScript) frontend and FastAPI + SQLite backend; containerized via Docker multi-stage builds and deployed to production on Fly.io with HTTPS via Let's Encrypt.
- Developed real-time metrics dashboard visualizing booking rates, revenue, carrier sentiment analysis, and negotiation outcomes; implemented autonomous negotiation engine with broker-configurable rules and margin protection, enabling 24/7 unattended carrier call handling.

Enterprise Nervous System

- Engineered a multi-agent incident response platform using AG2 swarms and MCP, orchestrating parallel retrieval across logs, GitHub, and Jira to automate root-cause analysis and postmortem generation in <19s per incident.
- Built an evaluation and monitoring pipeline spanning 21 production-style failure scenarios with deterministic scoring for evidence quality, actionability, reliability, citation integrity, and PII compliance, achieving 80% performance.

AgentProbe

- Built an automated adversarial evaluation platform for agentic AI systems using a distributed four-agent pipeline (Recon, Attack, Evaluator, Reporter) to identify prompt injection, PII leakage, and policy violations.
- Designed a hybrid evaluation engine combining deterministic rules, regex pipelines, and LLM-based scoring aligned with OWASP LLM risk categories for scalable automated assessment.

SKILLS

Languages & Systems	Python, C/C++, SQL, JavaScript/TypeScript, React, Docker, Git, SLURM
Machine Learning	PyTorch, LoRA/QLoRA, retrieval systems, model evaluation pipelines
AI / Agent Stack	LangGraph, CrewAI, AG2, MCP, RAG, vector database